

Dynamic Planet

Do Not Write on the Test

Time: 50 minutes

Multiple choice: 2 points, Written response: 5 points, Matching: 1 point each

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1. **What is Manning's Formula for discharge?**

2. **How much of the world's freshwater is concentrated in the great lakes?**
    a. 30%
    b. 22%
    c. 24%
    d. 21%

3. **Which of the following correctly describes a Meromictic Lake?**
    A. A lake that has uniform temperature from top to bottom at least once a year
    B. A lake which has layers of water that do not mix
    C. A lake that has layers mix at least once per year
    D. None of the above

4. **Which are factor(s) that determine when waves will break?**
    A. Climate
    B. Slope to the ocean floor
    C. Which zone the wave is in
    D. B and C

5. **___is the most common form of vertical structure**

6. **What is the Ca^+2 ion balance percentage for freshwater?**
    A. 17%
    B. 63%
    C. 68%
    D. 23%

7. **What percentage of freshwater is stored in swamps/marshes?**
    A. 0.03%
    B. 0.02%
    C. 0.01%
    D. 0.04%

8. **___ Lakes are lakes that have a high concentration of Ca^+2 and Mg^+2**

9. **Which of the following is an example of lithification?**
    A. Colonial Vertebrates
    B. Algae
    C. Sedimentary
    D. None of the above

10. **Which of the following is/are true?**
    A. Swamps and Marshes have acidic water
    B. Bogs, Marshes, and Swamps are the same type of wetland with just differing locations
    C. Bogs have acidic groundwater
    D. A and B

11. True or False: The condition of a wetland is completely dependent on the groundwater's quality

12. Identify a bog:

A. **B.** **C.** **D.**



13. Canyons form where the rate of stream ____ is faster than the mass wasting of the rock lining the walls along the sides of the channel.

14. Which of the following is an effect of dam construction?

A. Disturbance of reproduction cycles of surrounding species
B. Flooding
C. Altered recreational activities
D. A and C

15. Which of the statements below is true?

A. The intensity, inches/hour, of mist is 0.005 in/hour
B. Moderate rain is 50 rain drops per second per square foot
C. Water makes up of 99% of the earth
D. None of the above

16. Which of the following are causes of a patterned ground?

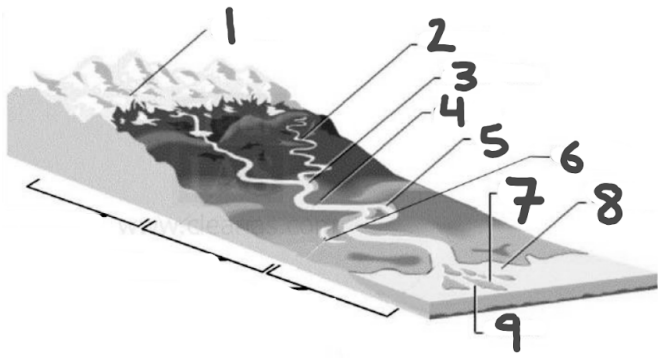
A. Rock Domain
B. Frost Heaves
C. None of the above
D. A and B

17. Which of the following are type(s) of floods?

A. Areal
B. Shoreline
C. Burst
D. A and B

18. What is a Seechi Disk?

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Matching: Place the correct term in the correct spot, each term may be used more than once: source, tributary, levee, delta, estuary, mouth, confluence, meander, oxbow lake

1.
2.
3.
4.
5.
6.
7.
8.
9.

19. Which of the following are formed through meander bend cutoffs, channel avulsions (abandonment of an entire channel segment), or mid-channel deposition and subsequent vegetation?

A. Anabranching Rivers
B. Straight Rivers
C. Meandering Rivers
D. Braided Rivers

20. _____ is a process where particles loosened by rain-splash erosion are transported by runoff water down the slope of a surface.

A. Rain Splash Erosion
B. Sheet Erosion
C. Rill Erosion
D. Gullys

21. Names at least 5 sources of water pollution (1 pt each).

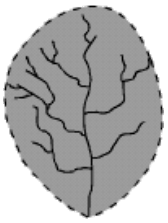
22. Name the three main steps of transpiration (2 pts each).

23. Transpiration is an invisible but important process. Choose the correct statement below:

A. A growing leaf will transpire many times more water than its own weight
B. An acre of corn gives off 5000 gallons of water each day
C. A fully grown pine tree transpires 7000 gallons of water each year
D. Transpiration is immeasurable

24. Name the soil moisture regime of the midwest.

25. What type of drainage pattern is this:



a. Parallel Drainage
b. Rectangular drainage patterns
c. Contorted patterns
d. Dendritic Drainage

26. True or False: Darcy's law can be applied in laminar flow

a. True
b. False

27. Which channel has a higher sinuosity?

a. Straight
b. Meandering with many bends
c. Meandering with few bends
d. Meandering with many cutoffs

28. The ground below the water table contains ____.

a. Zone of aeration
b. Zone of saturation
c. Zone of permeability
d. Zone of porosity

29. Karst features form from which type of rock?

a. Limestone
b. Marble
c. Granite
d. Gneiss

30. Glacial lakes form

a. when a glacier retreats and meltwater fills the hole left behind.
b. when a glacier forms and fills the hole.
c. when a glacier retreats and the empty hole is left behind.
d. when a glacier forms and creates a basin.

31. The combination of a second and third order stream produces

a. First order
b. Second order
c. Third order
d. Fourth order

32. What is a watershed?

a. an area or ridge of land that separates waters flowing to different rivers, basins, or seas
b. rivers that separate the land
c. a large body of water surrounded by land
d. a river or stream flowing into a larger river or lake

33. What is the difference between porosity and permeability in rocks?

a. Porosity is the percentage of void space in a rock, while permeability is percentage of rock space
b. Porosity is the ability of fluids to move through rocks, while permeability is the percentage of void space in a rock
c. Porosity is the percentage of rock space, while permeability is percentage of void space in a rock
d. Porosity is the percentage of void space in a rock, while permeability is the ability of fluids to move through rocks

34. How do sinkholes form?

a. When the land surface above collapses or sinks into the cavities
b. When the wind blows away the soil and rock
c. When the surrounding land is lifted above the sinkhole
d. When a river stops flowing and leaves a depression

35. The hydraulic gradient can be thought of as the

a. Depth of the water table
b. The slope of the water table
c. The length of the water table
d. The permeability of the water table

36. Leakance describes

a. Infiltration of water deep underground
b. The movement of water in confined aquifers to the ocean
c. The exchange of water between lakes and groundwater
d. The exchange of water between confined and unconfined aquifers

37. Which type of valley is characterised by steep sides that curve at the base of the valley wall?

A. V-Shaped Valley
B. Glacial Troughs
C. Flat-Floored Valley
D. S-Shaped Valley

38. Which of the following has coarser and denser particles that stay on the bed of the stream but later move because of saltation?

A. Dissolved Load
B. Bed Load
C. Suspended Load
D. None of the above

39. What is the degree of turbulence determined by?

40. What are the two initial stages of formation of caves (2.5 pts each)?

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Match the term with the definition:

A. **Gradient**
B. **Base level**
C. **Dynamic equilibrium**
D. **Nick points**
E. **Waterfalls**
F. **Stream capture**
G. **Deltas**
H. **Fans**

Definitions

41. ___ wetlands that form as rivers empty their water and sediment into another body of water.

42. ___ a measure of how steeply it loses height.

43. ___ part of a river or channel where there is a sharp change in channel slope.

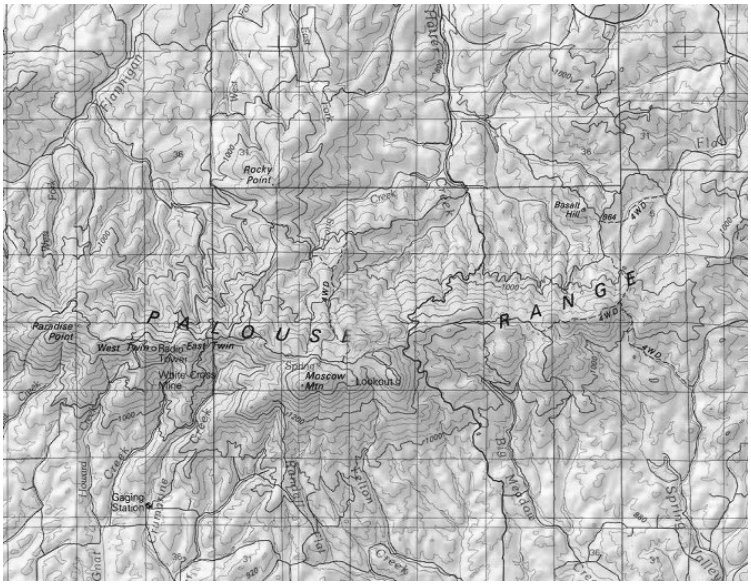
44. ___ a point in a river where water flows over a steep drop that is close to or directly vertical.

45. ___ when the net effect of the flowing water is neither erosion of the bed nor deposition of sediment.

46. ___ the imaginary horizontal level or surface to which sub-aerial erosion proceeds.

47. ___ a triangle-shaped deposit of gravel, sand, and even smaller pieces of sediment, such as silt.

48. ___ when an actively eroding low level stream encroaches on the drainage of a nearby stream flowing at a higher level and diverts part of the water of the higher stream.



49. Which direction does Big Meadow flow?

50. Calculate how much water flows through a stream on average in L/hour, if 100. cm of rainfall each year for a watershed of 100. m^2. Assume runoff is 50% of rainfall.